Name:_____

Exam Question Booklets
OCR Physics A

Booklet 11
12 Questions
96 marks

Things to remember:

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✓

✓

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1 Fig. 4.1 shows, at a given instant, the surface of the water in a ripple tank when plane water waves are travelling from left to right.

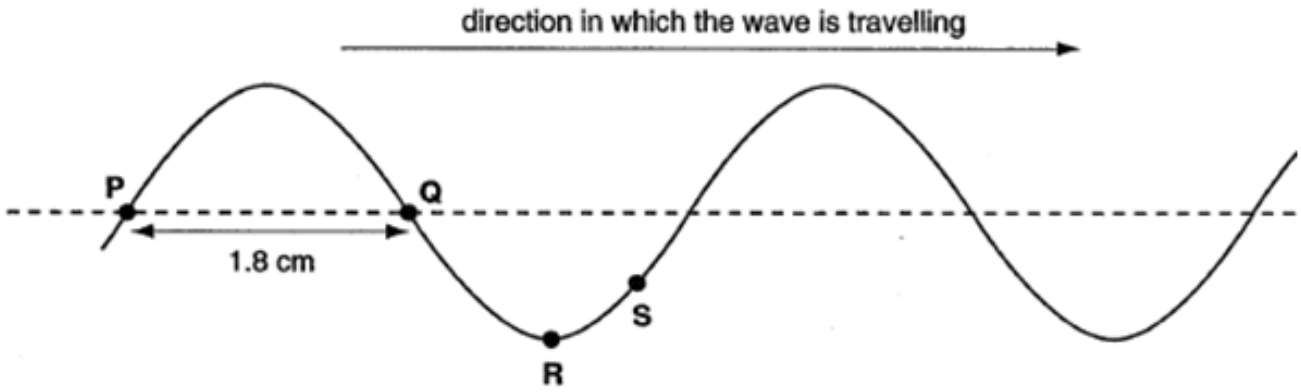

Fig. 4.1

(a) Show on Fig. 4.1

(i) the amplitude of the wave – label this **A** [1]

(ii) the wavelength – label this **λ**. [1]

(b) On Fig. 4.1

(i) draw the position of the wave a short time, about one-tenth of a period, later [2]

(ii) draw arrows to show the directions in which the particles at **Q** and **S** are moving during this short time. [2]

(c) State the phase difference between the movement of particles at **P** and **Q**.

phase difference = ...° [1]

(d) The frequency of the wave is 25 Hz and the distance between **P** and **Q** is 1.8 cm. Calculate

(i) the period of the wave

period = ...s [2]

(ii) the speed of the wave.

speed = ...m s^{-1} [3]

(a) **(i)** Both electromagnetic waves and sound waves can be **reflected**. State **two** other wave phenomena that apply to both electromagnetic waves and sound waves.

1. .....

2. ..... **[2]**

(ii) Explain why electromagnetic waves can be polarised but sound waves cannot be polarised.

.....

..... **[1]**

(iii) Describe briefly an experiment to demonstrate the polarisation of microwaves in the laboratory.

In your answer you should make clear how your observations demonstrate polarisation.

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..... **[4]**

(a) Electric current is the flow of charged particles. Name the charged particle responsible for electric current in

(i) a metal,

..... [1]

(ii) an electrolyte (conducting solution).

..... [1]

(b) Name the two quantities required to calculate the electrical resistance of a component in an electric circuit.

1. .....

2. ..... [2]

(c) A digital watch uses a 1.3 V cell. It delivers a charge of 650 C at a constant rate during its lifetime of 1.6×10^7 s. Calculate

(i) the current delivered by the cell,

current = ... A [3]

(ii) the total number of electrons passing through the cell during its lifetime of 1.6×10^7 s.

number = ... [2]

4

(a) Define the *kilowatt-hour* (kW h).

.....

.....[1]

(b) On average, a student uses a computer of power rating 110 W for 4.0 hours every day. The computer draws a current of 0.48 A and its screen emits visible light of average wavelength 5.5×10^{-7} m.

(i) For a period of **one** week, calculate

1. the number of kilowatt-hours supplied to the computer

number of kW h = ...[2]

2. the cost of operating the computer.
(The cost of each kW h is 7.5p)

cost = ... p [1]

5

A 120 W filament lamp transforms 5.8 kW h. Calculate the time in seconds for which the lamp is operated.

time = ... s [2]

6

(a) Thermistors and light-dependent resistors are useful components in potential divider circuits.

(i) Draw the symbol for a thermistor and describe how the resistance of a negative temperature coefficient (NTC) thermistor is affected by temperature.

.....

.....

..... [2]

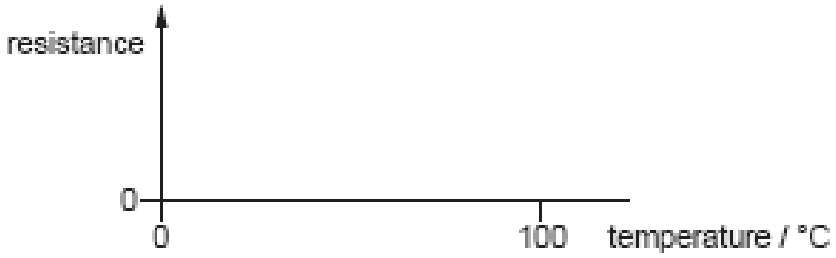
(ii) Draw the symbol for a light-dependent resistor and describe how its resistance is affected by the intensity of light.

.....

.....

..... [2]

(b) **(i)** On the axes below, show how the resistance of a pure metal conductor in the form of a wire depends on its temperature.



[1]

(ii) In this question two marks are available for the quality of written communication.

Describe an experiment to determine how the resistance of a wire varies with its temperature. Assume that no ohmmeter is available.

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.....[5]

(a) Define electrical resistance.

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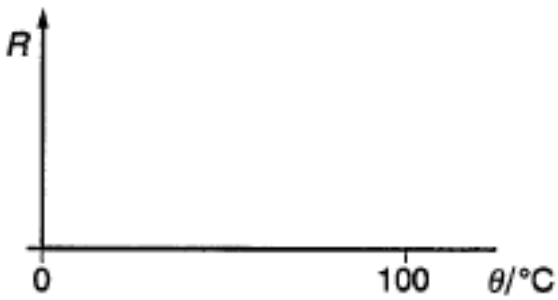
..... [2]

(b) With the aid of a sketch graph, describe how the resistance R of a negative temperature coefficient (NTC) thermistor changes with temperature θ.



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..... [2]

(c) Fig. 2.1 shows the I/V characteristic of a tungsten filament lamp.

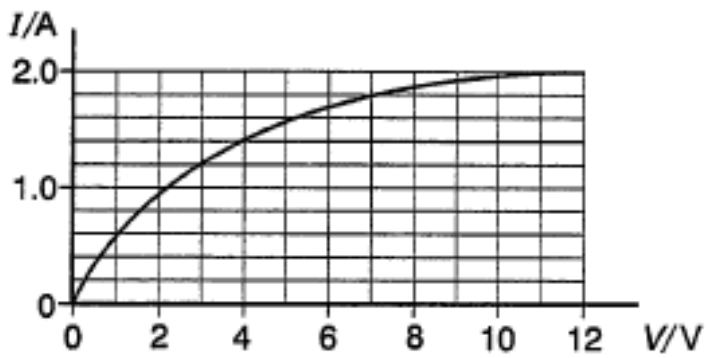

Fig. 2.1

(i) State how, and explain why, the resistance of the filament lamp changes as the potential difference V across it increases.

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..... [2]

(ii) A 5.0 Ω resistor and the tungsten filament lamp are connected in series to a d.c. power supply of e.m.f. 24 V. The current drawn from the power supply is 2.0 A.

1. Calculate the total power delivered by the supply.

power ..... W [2]

2. Use **Fig. 2.1** to determine the resistance of the filament lamp when the current in it is 2.0 A.

resistance = ... Ω [2]

In this question, one mark is available for the quality of written communication.

A simple electrical thermometer circuit used to monitor the temperature of a water bath is shown in Fig. 3.2.

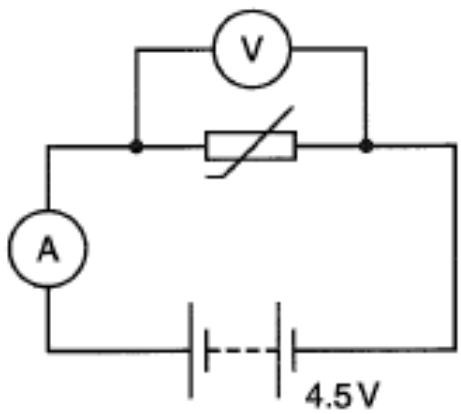

Fig. 3.2

The ammeter and the battery both have negligible internal resistances. The voltmeter connected across the negative temperature coefficient (NTC) thermistor has an infinite resistance.

Discuss how each meter responds to an increase in the temperature of the water.

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.....[4]

(a) State the difference between the directions of conventional current and electron flow.

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.....[1]

(b) In the space provided below, draw the symbol for a light-dependent resistor (LDR) and state how its resistance is affected by the intensity of light falling on it.

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.....

.....[2]

(c) State Ohm's law.

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.....[2]

(d) Current against voltage (*I*/*V*) characteristics are shown in Fig. 1.1a for a metallic conductor at a constant temperature and in Fig. 1.1b for a particular thermistor.

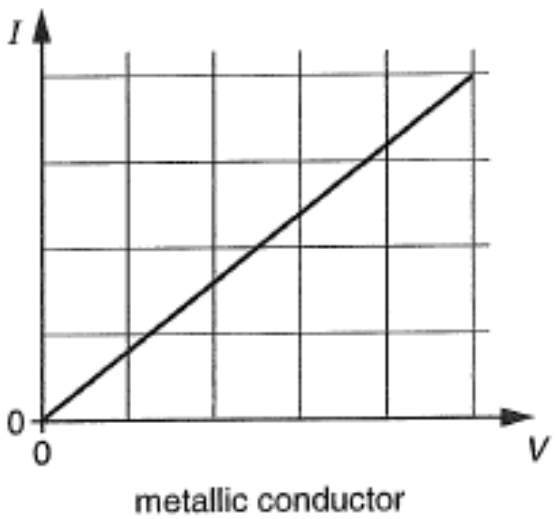

Fig. 1.1a

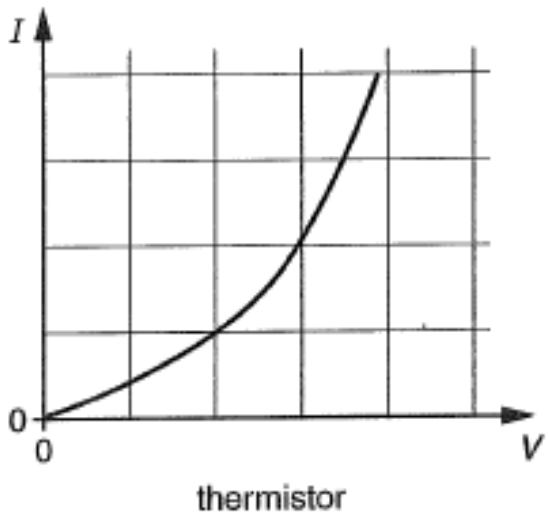

Fig. 1.1b

(i) Sketch the variation of resistance R with voltage V for

1. the metallic conductor at constant temperature (draw this on Fig. 1.2a)

2. the thermistor (draw this on Fig. 1.2b). [3]

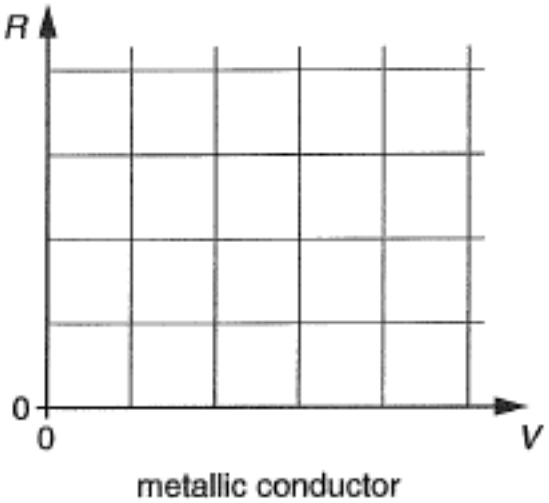

Fig. 1.2a

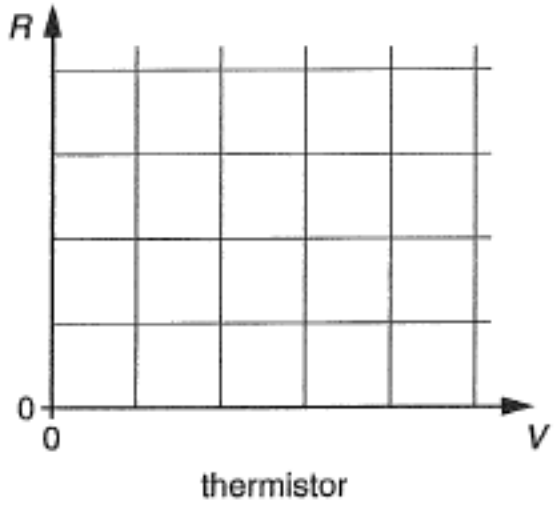

Fig. 1.2b

(ii) State and explain the change, if any, to the graph of resistance against voltage for the metallic conductor

1. when the temperature of the metallic conductor is kept constant at a **higher** temperature

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2. when the length of the conductor is doubled but the material, temperature and the cross-sectional area of the conductor remain the same.

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.....[4]

(a) **(i)** Explain with the aid of diagrams how the principle of superposition accounts for the constructive and destructive interference of waves.

1 constructive interference

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.....

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.....

2 destructive interference

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.....

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[4]

(ii) In order to produce an observable interference pattern the wave sources must be coherent. Explain the meaning of coherence.

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.....[2]

(b) Fig. 3.1 shows a plan view of an experiment to demonstrate interference of microwaves. **T** is a microwave transmitter and **D** is a detector.

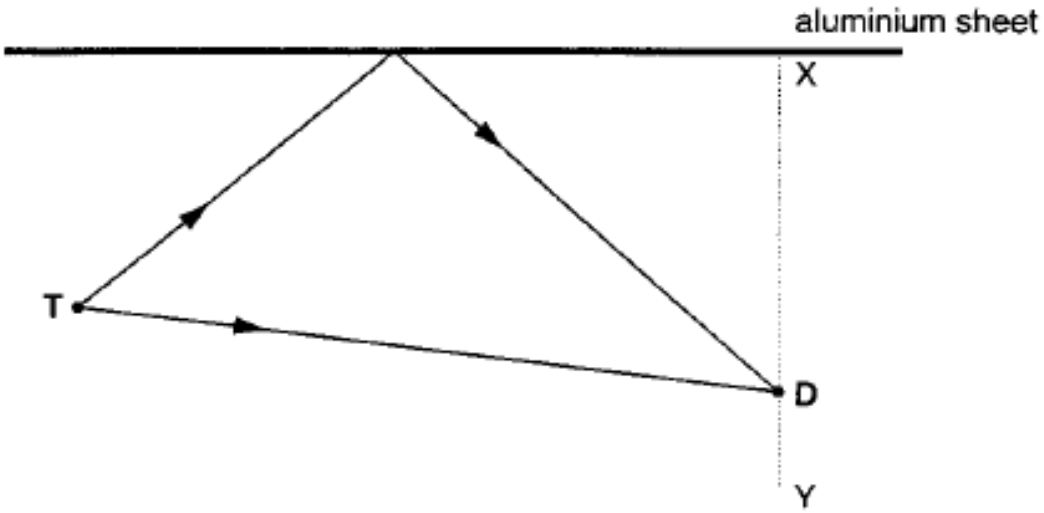

Fig. 3.1

Microwaves from **T** reach the detector **D** by two routes. Some travel directly to **D** in a straight line and others reach **D** after being reflected by the large aluminium sheet. When the detector is moved along the line XY perpendicular to the sheet the detector registers a sequence of maxima and minima.

(i) By referring to the path difference between the two sets of waves arriving at **D**, explain why maxima and minima are formed.

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.....

.....

.....[2]

(ii) Describe and explain how the separation between neighbouring maxima would change if microwaves with a shorter wavelength were used.

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.....[2]

11

Describe an experiment to determine the wavelength of monochromatic light (i.e. light of one wavelength) using a double-slit of known slit separation.

Include in your answer

- a labelled diagram showing how the apparatus is arranged
- a list of the measurements required to determine the wavelength λ of the light
- the formula, with all symbols identified, that could be used to determine λ.

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.....[6]

(a) Explain what is meant by the *principle of superposition* of two waves.

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..... [2]

(b) In an experiment to produce an observable interference pattern, two monochromatic light sources, S_1 and S_2, are placed in front of a screen, as shown in Fig. 5.1.

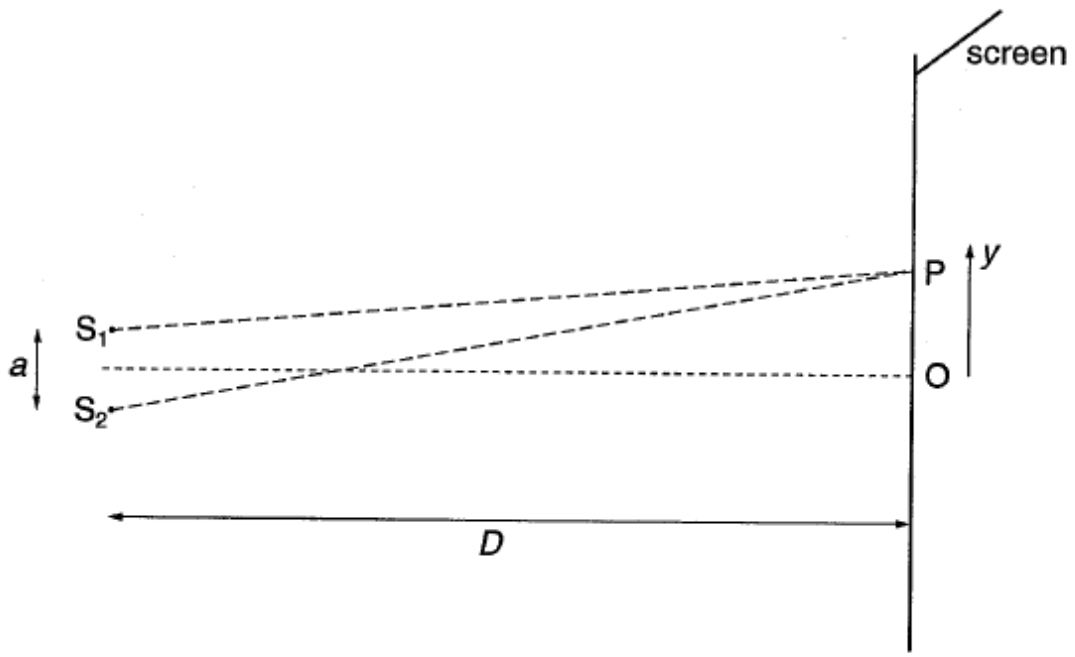

Fig. 5.1

(i) In order to produce a clear interference pattern on the screen the light sources must be *coherent*. State what is meant by *coherence*.

.....

.....

..... [2]

(ii) The point P is the first position of *minimum* intensity moving up from the central point of maximum intensity O. State, in terms of the wavelength λ, the magnitude of the path difference between S_1P and S_2P.

..... [1]

(iii) When $D = 1.5\,\text{m}$ and light of wavelength $6.4 \times 10^{-7}\,\text{m}$ is incident on the double slit, the distance between P and O is 4.0 mm. Calculate the slit separation a.

a = ... m [3]

(iv) Sketch on the axes of Fig. 5.2 the variation of the intensity of the light on the screen with distance y from O. [2]

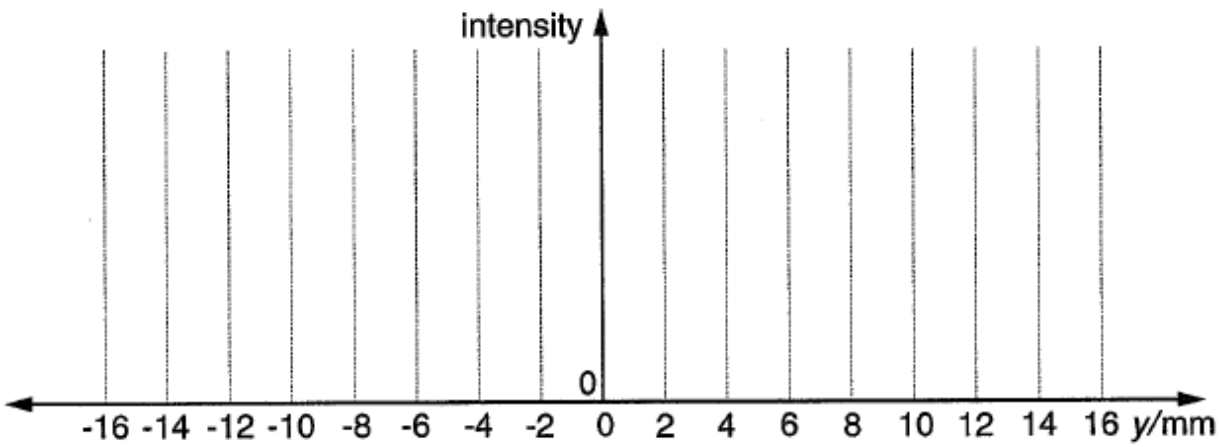

Fig. 5.2